Diffferential Geometry

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Abstract. Differential Geometry is regarded as a branch of mathematics that uses cacculus and linear algebra to study the shapes, curves, and surfaces of smooth curved spaces. Unlike elementary geometry, which studies flat surfaces and rigid shapes, differential geometry focuses on local and global characteristics of curved spaces where geometric properties vary continuously from point to point. This framework serves as the mathematical language for modern theoretical physics.

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1. Chapter : Curves

1.1. A Brief Introduction

Differential Geometry can be broken into two types:

1. **Classical** – deals with studying local properties of curves and surfaces. By *local* properties, we are dealing with properties that depend only on the curve/surface in small neighbourhoods.

2. **Global** – deals with how local properties may influence the entire curve.

1.2. Parameterised Curves

Note that $R^3 = \mathbf{E^3} = (x, y, z)$. This tripled order is a subset of R^3, as previously discussed in chapter 0. One way we can characterise curves is by whether or not it is differentiable.

Definition 1.1. A real function f of a real variable x is **differentiable, or smooth** if it has derivatives at all points. This means no sharp turns or edges.

Definition 1.2. A **parameterised differentiable curve** is a differentiable map: $\alpha : I \to R^3$ of an open interval $I = (a, b)$ of the real line R into R^3

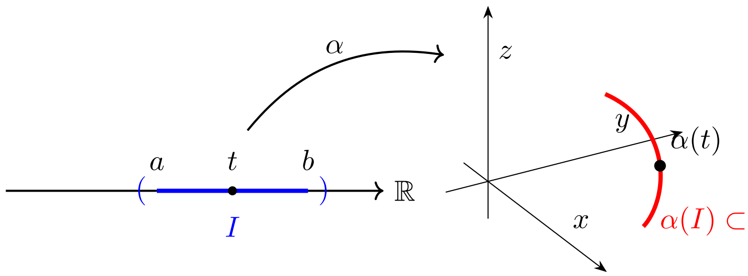

Figure 1: Visualization of a parameterised differentiable curve mapping an open interval I into $\mathbb{R}^3$.

In definition 1.2, differentiable means α is a correspondence, or *function* which maps each $t \in I$ into a point $\alpha(t) = (x(t), y(t), z(t)) \in R^3$ such that $x(t), y(t), z(t)$ are differentiable. t is called the *parametre* of the curve. If we denote $x'(t)$ as the first derivative of x at a point t, then the vector

$$\alpha'(t) = (x'(t), y'(t), z'(t)) \in R^3 \tag{1}$$

is the **tangent vector**.

Moving forth, recall some vector properties. Let $u = u_1, u_2, u_3 \in R^3$. Its length, or *norm*, is defined as:

$$|u| = \sqrt{u_1^2 + u_2^2 + u_3^2} \tag{2}$$

Using the previously defined u as one vector and v as another vector, then the inner product is defined as:

$$u \cdot v = |u||v| \cos(\theta) \tag{3}$$

Additionally, note that the following properties hold:

1. If u and v are non-zero vectors, $u \cdot v = 0$ is true iff u is orthogonal to v.
2. $u \cdot v = v \cdot u$
3. $\lambda(u \cdot v) = \lambda u \cdot v = u \lambda v$
4. $u(v + w) = ()u \cdot v) + (u \cdot w)$

Using the natural frame fields in chapter 0, we find the expression for the inner product such that:

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3 \tag{4}$$

and the derivative:

$$\frac{d}{dt}(u(t) \cdot v(t)) = u'(t) \cdot v(t) + u(t) \cdot v'(t). \tag{5}$$

1.3. Regular Curves; Arc Length

Recall that $\alpha : I \to R^3$ is a parameterised differentiable curve. For each $t \in I$ where $\alpha'(t) \neq 0$, there is a straight line which confirms both α and its derivative, where $\alpha'(t)$ is some vector. This is known as a **tangent line**. Any point where $\alpha'(t) = 0$ is called a **singular point**.

Definition 1.3. A parameterised differentiable curve $\alpha : I \to R^3$ is said to be **regular** if $\alpha'(t) \neq 0 \forall t \in I$.

Given $t \in I$, the arc length, by definition, is:

$$s(t) = \int_{t_0}^{t} |\alpha'(t)|\, dt, \tag{6}$$

where

$$|\alpha'(t)| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} \tag{7}$$

If the parametre t is already the arc length, $\frac{ds}{dt} = 1 = |\alpha'(t)|$. If that is the case, then:

$$s = \int_{t_0}^{t} dt = t - t_0; \tag{8}$$

Given the curve α parameterised by arc length $s \in (a, b)$, we can consider the curve $\beta \in (-b, -a)$ by $\beta(-s) = \alpha(s)$, which has the same trace but described in the opposite direction. We say that these two curves differ by a **change of orientation**.

1.4. Curves: Summary

Recall the velocity vector:

$$\alpha'(t) = \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t)\right). \tag{9}$$

The speed for some given α is defined by:

$$v = \|\alpha'\| = \left(\left(\frac{d\alpha_1}{dt}\right)^2 + \left(\frac{d\alpha_2}{dt}\right)^2 + \left(\frac{d\alpha_3}{dt}\right)^2\right)^{1/2}. \tag{10}$$

In physics, the distance travelled by a particle in motion is determined by integrating its speed with respect to time. The definition of arc length is then:

$$\int_a^b \|\alpha'(t)\|\, dt. \tag{11}$$

Sometimes, we aren't interested in the speed of the curve but only the path. In order to separate speed from the path of the curve, we use *re–parameterisation*. This is achieved by reparameterising a curve β which has a unit speed $||\beta'|| = 1$.

Theorem 1.4. *If α is a regular curve in $\mathbf{E}^3$, then there exists a reparametrisation β of α such that β has unit speed.*

A reparametrisation $\alpha(h)$ of a curve α is said to be *orientation-preserving* if $h' \geq 0$, *orientation-reversing* if $h' \leq 0$.

Now, we adapt the notion of a *vector field* to the study of curves.

Definition 1.5. A vector field Y on a curve $\alpha : I \to \mathbf{E}^3$ is a function that assigns to each number t in I a tangent vector $Y(t)$ to $\mathbf{E}^3$ at the point $\alpha(t)$.

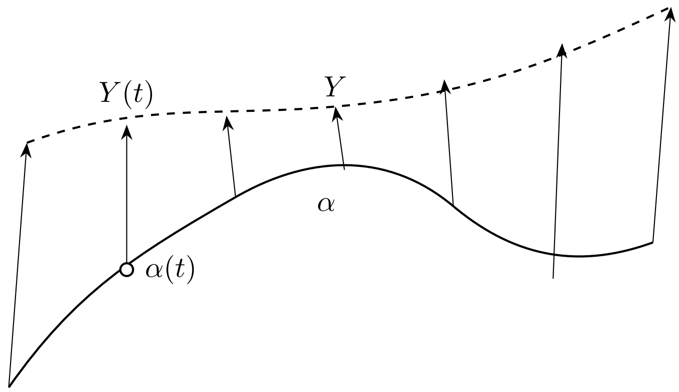

Figure 2: A vector field Y along a curve α.

One important note is that the properties of vector fields on curves are analogous to those of vector fields on $\mathbf{E^3}$. To differentiate a vector field on α simply differentiates its Euclidean coordinate functions, thus giving a new vector field on α.

The second derivative of α is α'', which we may know as *acceleration*.

$$\alpha'' = \left(\frac{d^2\alpha_1}{dt^2}, \frac{d^2\alpha_1}{dt^2}, \frac{d^2\alpha_3}{dt^2}\right)_\alpha . \tag{12}$$

1.5. The Vector Product in R^3

This section covers the notion of a vector space. Two ordered bases $e = e_i, f = f_i$ of a n–dimensional vector space V have the same orientation if the matrix change of basis has a positive determinant ($det(M) > 0$). This is denoted as $e \sim f$. Note that the following are also satisfied:

1. $e \sim e$.
2. If $e \sim f$, then $f \sim e$.
3. If $e \sim f$, $f \sim g$, then $e \sim g$.

Each of the equivalence classes determined by the above relations is called the **orientation** of V. $\therefore V$ has two equivalence classes – positive or negative.

Definition 1.6. Let $u, v \in \mathbb{R}^3$. The vector product of u and v (in that order) is the unique vector $u \wedge v \in \mathbb{R}^3$ characterized by

$$(u \wedge v) \cdot w = \det(u, v, w) \quad \text{for all } w \in \mathbb{R}^3. \tag{13}$$

Here $\det(u, v, w)$ means that if we express u, v, and w in the natural basis $\{e_i\}$,

$$u = \sum u_i e_i, \quad v = \sum v_i e_i, \quad w = \sum w_i e_i, \quad i = 1, 2, 3, \tag{14}$$

then

$$\det(u, v, w) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}. \tag{15}$$

Note that the use of the wedge also means *cross product*. With the above definition, the following are satisfied:

1. $u \wedge v = -v \wedge u$ (anticommutativity).

2. $u \wedge v$ depends linearly on u and v; i.e., for any real numbers a, b, we have

$$(au + bw) \wedge v = au \wedge v + bw \wedge v.$$

3. $u \wedge v = 0$ if and only if u and v are linearly dependent.

4. $(u \wedge v) \cdot u = 0$, $(u \wedge v) \cdot v = 0$.

Proof. We want to understand the geometric meaning of the vector product $u \wedge v$. To do this, we use a general identity that links dot products and vector products for any four vectors u, v, x, y:

$$(u \wedge v) \cdot (x \wedge y) = \begin{vmatrix} u \cdot x & v \cdot x \\ u \cdot y & v \cdot y \end{vmatrix} \tag{16}$$

Because both sides of this equation change linearly if we scale or add vectors, we only need to test this on the standard basis vectors $\{e_1, e_2, e_3\}$ to prove it works universally:

$$(e_i \wedge e_j) \cdot (e_k \wedge e_l) = \begin{vmatrix} e_i \cdot e_k & e_j \cdot e_k \\ e_i \cdot e_l & e_j \cdot e_l \end{vmatrix} \tag{17}$$

Now, let us look at a special case. If we pick $x = u$ and $y = v$, the left side becomes the squared length of the vector product, $|u \wedge v|^2$. Evaluating the determinant on the right side gives:

$$|u \wedge v|^2 = \begin{vmatrix} u \cdot u & u \cdot v \\ u \cdot v & v \cdot v \end{vmatrix} = (u \cdot u)(v \cdot v) - (u \cdot v)^2 \tag{18}$$

We can simplify this using basic geometry. We know that $u \cdot u = |u|^2$, $v \cdot v = |v|^2$, and $u \cdot v = |u||v| \cos\theta$, where θ is the angle between the two vectors. Substituting these in:

$$|u \wedge v|^2 = |u|^2|v|^2 - (|u||v| \cos\theta)^2 = |u|^2|v|^2(1 - \cos^2\theta) \tag{19}$$

Using the identity $1 - \cos^2\theta = \sin^2\theta$, we get:

$$|u \wedge v|^2 = |u|^2|v|^2 \sin^2\theta = A^2 \tag{20}$$

Taking the square root shows that the length $|u \wedge v| = |u||v| \sin\theta$ is precisely the area A of the parallelogram spanned by u and v.

In summary, $u \wedge v$ is a vector that stands completely perpendicular to the flat plane containing u and v. Its length matches the area of their shared parallelogram, and its upward direction follows the right-hand rule, meaning $\{u, v, u \wedge v\}$ forms a positively oriented basis. $\square$

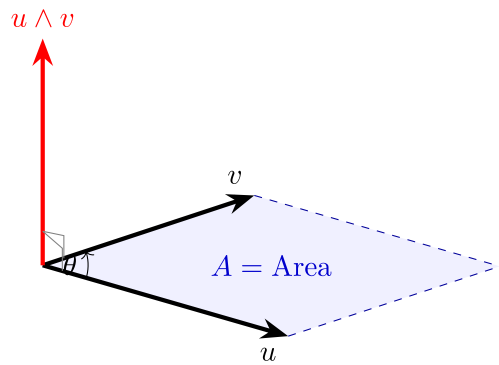

Figure 3: Geometric representation of the vector product $u \wedge v$ acting perpendicular to the plane with a norm equal to the parallelogram area A.

The vector product is not associative:

$$(u \wedge v) \wedge w = (u \cdot w)v - (v \cdot w)u, \tag{21}$$

At this point, one should be able to grasp that if we let u and v represent differentiable maps from the interval (a, b) to R^3, $t \in (a, b)$ if it follows that $u)t) \wedge v(t)$ is differentiable such that:

$$\frac{d}{dt}(u(t) \wedge v(t)) = \frac{du}{dt} \wedge v(t) + u(t) \wedge \frac{dv}{dt}. \tag{22}$$

1.6. The Local Theory of Curves Parameterised by Arc Length

This section has important results regarding curves. Suppose a curve is parameterized by its arc length s, denoted as $\alpha : I = (a, b) \to \mathbb{R}^3$. Because it is a unit-speed curve, its velocity vector satisfies $\|\alpha'(s)\| = 1$. Consequently, the magnitude of the acceleration vector, $\|\alpha''(s)\|$, quantifies how quickly the tangent vector changes direction relative to an adjacent point. Geometrically, this norm reflects the rate at which the curve deviates or pulls away from its local tangent line near s.

Definition 1.7. Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length $s \in I$. The number $\|\alpha''(s)\| = \kappa(s)$ is called the curvature of α at s.

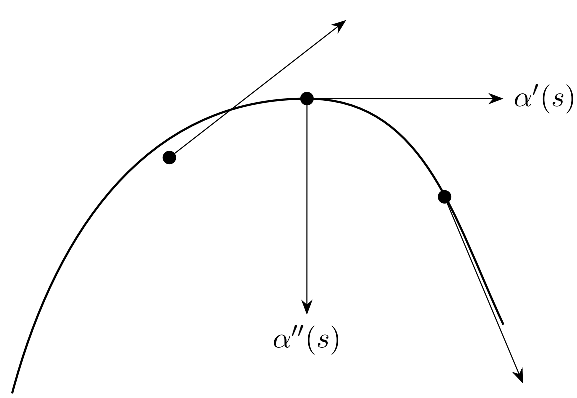

Figure 4: Velocity $\alpha'(s)$ and acceleration $\alpha''(s)$ vectors along a unit-speed curve.

From fig. 4, $\Delta\theta$ is found by $|\alpha''(t)|$. As we travel from one point to the next, the arrows become tilted in some degree; thus, the tangent vector must rotate to stay in alignment with the curve.

At $\kappa \neq 0$, we apply a unit vector $n(s)$. Therefore:

$$\alpha''(s) = \kappa(s)n(s) \tag{23}$$

Since $\alpha''(s) \perp \alpha'(s)$, both $\alpha''(s)$ and $\alpha'(s)$ are orthogonal. Recall that:

$$\frac{d}{ds}(\alpha'(s) \cdot \alpha'(s)) = \frac{d}{ds}(1) \tag{24}$$

and the dot product $\alpha''(s) \cdot \alpha'(s) = 0$. Under this reasoning, we also note that $n(s) \perp \alpha'(s)$. The plane that forms between these two is called the *osculating plane*.

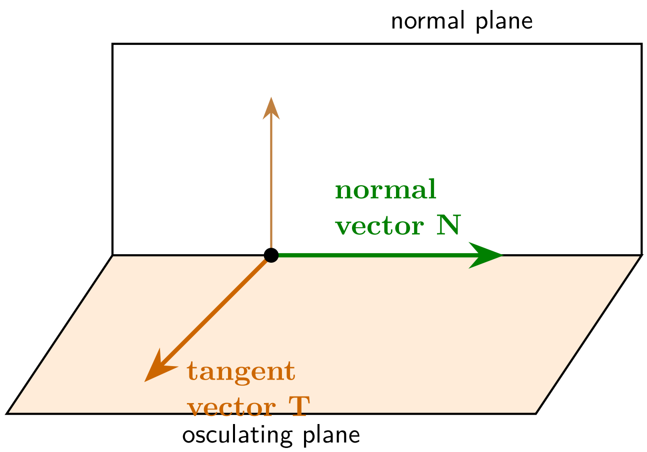

Figure 5: The Frenet-Serret vectors T and N spanning the shaded osculating plane.

When $\kappa(s) = 0$, then there is no need to define the osculating plane. Therefore, there is no need to define the normal vector either. These things in this case are *undefined*.

To note, we are saying that $n(s)$ is heavily reliant on $\kappa(s)$. Hence, the osculating plane is heavily reliant on $\kappa(s)$.

Going forth, we let $t(s) = \alpha'(s)$. Thus, $t'(s) = \alpha''(s) = n(s)\kappa(s)$. The unit vector $b(s) = t(s) \wedge n(s)$ is normal to the osculating plane and will be called the *binormal vector* at s.

Since it a unit vector to the plane, $b'(s)$ measures the rate of change of the neighbouring planes to the osculating plane at s.

$$\begin{aligned}
b'(s) &= \frac{d}{ds}[t(s) \wedge n(s)] \\
&= t'(s) \wedge n(s) + t(s) \wedge n'(s) && \text{(By Product Rule)} \\
&= [\kappa(s)n(s)] \wedge n(s) + t(s) \wedge n'(s) && \text{(Substitute } t'(s) = \kappa(s)n(s)\text{)} \\
&= \kappa(s)[n(s) \wedge n(s)] + t(s) \wedge n'(s) \\
&= \kappa(s)[0] + t(s) \wedge n'(s) && \text{(Since } n(s) \wedge n(s) = 0\text{)} \\
&= t(s) \wedge n'(s);
\end{aligned}$$

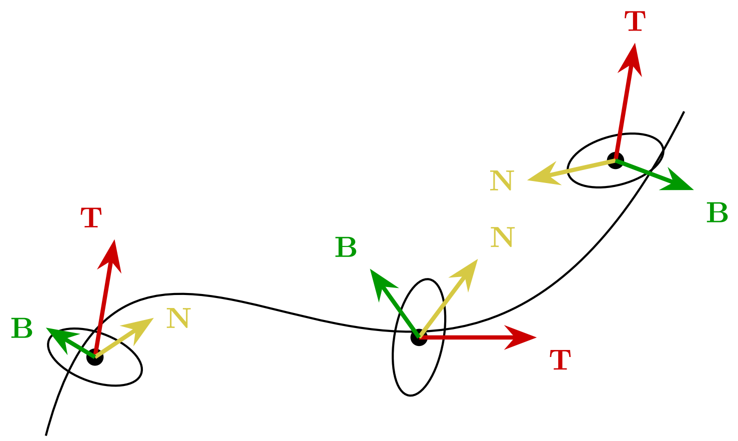

Figure 6: The moving Frenet-Serret frame $(\mathbf{T}, \mathbf{N}, \mathbf{B})$ along a space curve.

Definition 1.8. Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length s such that $\alpha''(s) \neq 0$, $s \in I$. The number $\tau(s)$ defined by $b'(s) = \tau(s)n(s)$ is called the torsion of α at s.

Definition 1.9. Plane Curves show all geometric points traced out by a curve lie entirely w/in a single flat 2D plane.

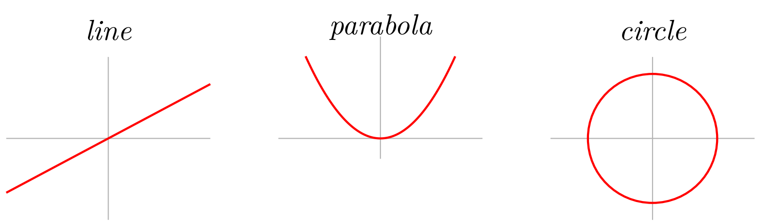

Figure 7: Classic examples of plane curves.

If α is a plane curve, then then plane curve is in agreement with the osculating plane such that they occupy the same plane. This means $\tau \equiv 0$.

The three orthogonal unit vectors we have just described are those that make up the *Frenet tihedron*.

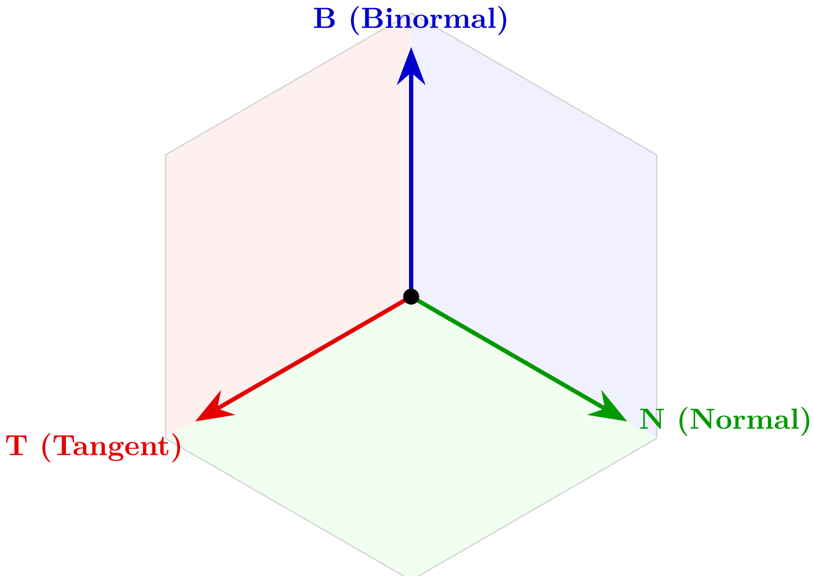

Figure 8: The Frenet Trihedron showing the orthogonal basis vectors **T**, **N**, and **B**.

What we didn't comput was $n'(s)$, so out of curiousity, we obtain:

$$\begin{aligned} n'(s) &= \frac{d}{ds}[b(s) \wedge t(s)] \\ &= b'(s) \wedge t(s) + b(s) \wedge t'(s) && \text{(By Product Rule)} \\ &= [\tau(s)n(s)] \wedge t(s) + b(s) \wedge [\kappa(s)n(s)] && \text{(Substitute } b' \text{ and } t'\text{)} \\ &= \tau(s)[n(s) \wedge t(s)] + \kappa(s)[b(s) \wedge n(s)] \\ &= \tau(s)[-b(s)] + \kappa(s)[-t(s)] && \text{(Using cyclic cross products)} \\ &= -\tau(s)b(s) - \kappa(s)t(s). \end{aligned}$$

These elucidate what we know as *Frenet formulas*:

$$\begin{cases} t' = kn, \\ n' = -kt - \tau b, \\ b' = \tau n \end{cases} \tag{25}$$

The use of the Frenet frame allows us to describe any curve as a result of bending and twisting.

This brings us to the fundamental theorem of the local theory of curves.

Theorem 1.10. *Given differentiable functions $\kappa(s) > 0$ and $\tau(s)$, $s \in I$, there exists a regular parametrized curve $\alpha : I \to \mathbb{R}^3$ such that s is the arc length, $\kappa(s)$ is the curvature, and $\tau(s)$ is the torsion of α.*

Moreover, any other curve $\tilde{\alpha}$ satisfying the same conditions differs from α by a rigid motion; that is, there exists an orthogonal linear map ρ of $\mathbb{R}^3$, with positive determinant, and a vector c such that $\tilde{\alpha} = \rho \circ \alpha + c$.

This tells us two things:

1. If you have a function w/ curvature $\kappa(s) > 0$ and a function with any torsion, a 3D cure with these exact properties exist.

2. If two different curves have the same κ and τ, then they are exactly identical minus where they are and how they are tilted.

   The proof (for II) can be explored intuitively. Say we have two bent wires on a table, one on the right and one on the left.

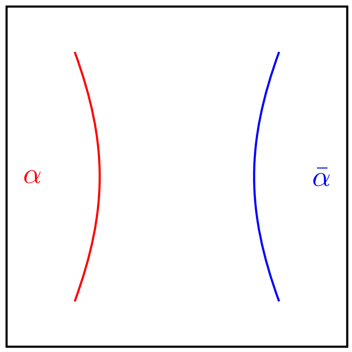

Figure 9: Geometric comparison between the regular curve α and its counterpart $\bar{\alpha}$.

to get $al\bar{p}ha$ to perfectly overlay α, we apply the theorem (or simply some rotation plus translation).

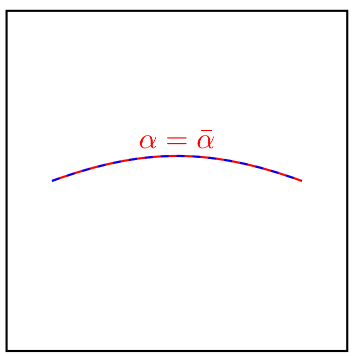

Figure 10: Coincident plane curves where α and $\bar{\alpha}$ lie exactly on top of each other.

Note that the sum of squares method is applied because we want to see the distance between the two wires are 0. This thells us they are locked together.

REFERENCES

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[2] Barrett O'Neill, *Elementary Differential Geometry*, Revised 2nd ed., Academic Press, 2006.